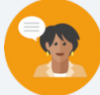


12.7 Deriving the Equation of a Circle

Lesson Introduction

 Benchmark	MA.912.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.		 Learning Targets	- I can use the Pythagorean Theorem to derive the equation of a circle. - I can determine key features of a circle given the graph or equation. - I can write the equation of a circle using key features.	
 Benchmark Coherence	Building On:			Working Toward:	
	MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.			N/A	
 Essential Questions	- How can you use the Pythagorean Theorem to derive the equation of a circle? - What are the two key features of a circle? - How can you write the equation of a circle when given the center and radius?				
 Lesson Narrative	This lesson activates prior knowledge regarding the Pythagorean Theorem. While using the Pythagorean Theorem, students will use the key features of a circle to derive the equation of a circle in the center-radius form. This will be done in the coordinate plane where students are given coordinates to find distance. Students will have the opportunity to use the formula derived to write the equation of a given circle on a coordinate plane.				
 Component Summary	12.7.1 Warm-Up (~5 min.)	Notice and wonder with Roundabouts (<i>SE p. 261</i>)	12.7.3 Guided Instruction (10-15 min.)	Using key features to write the equation of a circle (<i>SE p. 265</i>)	
	12.7.2 Exploration (25-30 min.)	Deriving the equation of a circle (<i>SE p. 261</i>)	12.7.4 Wrap-Up (~5 min.)	Ticket Out the Door determining key features and writing equation of circle (<i>SE p. 266</i>)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Center-Radius Form <u>Additional Terms:</u> - Pythagorean Theorem		N/A		The Pythagorean Theorem is used extensively in this lesson. Consider having reference cards available for struggling students who are having trouble with remembering the theorem.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle finding the length of the segment when given variables instead of numbers. - Students may have forgotten which part of the triangle are the legs and which part is the hypotenuse. - Students may believe that distance can be negative.	12.7.2 Exploration is meant to derive the equation of the circle, not necessarily find the actual length of a segment. - Pay close attention to the circles whose center coordinates are a combination of both positive and negative values and guide students through the process of connecting the equation of the circle to the sign of the numbers.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time 12.7.2 Exploration is structured as an interactive opportunity for students to revise and refine both their ideas and their verbal and written output. This routine allows students to ask clarifying questions, if needed, and continue on to the next question. Consider having students do this throughout the component, so students can get stronger with writing equations of a circle in standard form.	MA.K12.MTR.1.1: Participate in Effortful Learning 12.7.2 Exploration actively allows students to participate in effortful learning, both individually and collectively. Students will analyze the problems, ask questions that will help them find the equation of the circle, and help and support each other while attempting the next problems in the set. 12.7.2 Exploration will foster perseverance in students as they solve progressively more difficult problems.
 Differentiate Instruction	12.7.2 Exploration is set up to allow students to work through a series of questions that become progressively more complex. The diagrams of the circles referenced throughout 12.7.2 Exploration can assist student who need a visual of how the Pythagorean Theorem and writing the equation of a circle are related. There is also a lot of reading in this section. Provide an auditory entry-point from hearing the verbal description of the written representations.	
Lesson Notes:		

12.7.1 Warm-Up: Roundabouts

Component Overview: Students examine the concept of circles using a real-life example of roundabouts.



12.7.1 Warm-Up: Roundabouts

Roundabouts promote a continuous flow of traffic because drivers do not have to wait for a green light at a roundabout to get through the intersection. A diagram of a roundabout is shown.



How are tangent lines similar to the far-right lanes entering the roundabout?

How could circumference be used when finding the distance a car traveled through a roundabout?

1. What do you wonder about the diagram?

Answers may vary.

I wonder how traffic is controlled in a roundabout. I wonder why there are no cars in the inner circle of the roundabout.

2. How many different paths could the red car take?

The red car can only take one path because he is in the outer circle.

3. List at least two things you have studied in Geometry so far that may be helpful to know when constructing a roundabout.

Answers may vary.

What is the distance around the roundabout?

How many feet across is the roundabout?

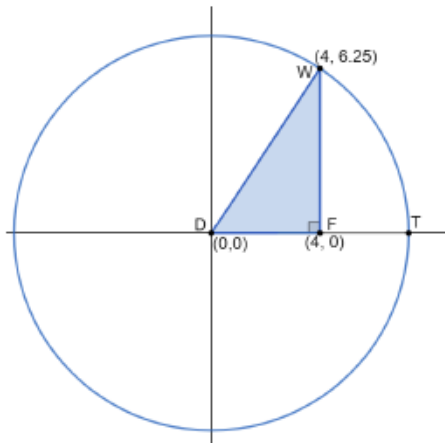
12.7.2 Exploration: Deriving the Equation of a Circle

Component Overview: Students use diagrams to derive the equation of a circle on the coordinate plane. Students use given coordinates and the Pythagorean Theorem to find the side lengths of the triangle created in a circle, which will lead students into writing the equation of a circle.



12.7.2 Exploration: Deriving the Equation of a Circle

1. The center of circle D is located at $(0,0)$. Point W lies on circle D at the point $(4, 6.25)$. Right triangle DFW is drawn inside the circle, such that DW is a circle's radius and point F , $(4,0)$, lies on radius DT .



- a. Find the length of each side.

Line Segment	Distance Between Endpoints
$\overline{DF}$	4
$\overline{FW}$	6.25

- b. Use the values you found for the lengths of $\overline{DF}$ and $\overline{FW}$ to complete the equation that can be used to determine the length of $\overline{DW}$ by applying the Pythagorean Theorem.

$$(DW)^2 = (4)^2 + (6.25)^2$$



Compare and Connect

Begin and end 12.7.2 Exploration by prompting students to make connections between what they know about circles, and what they know about their equations.

Before the beginning: What do you know about the equation of a circle? Take a guess – what do you think it looks like? Why?

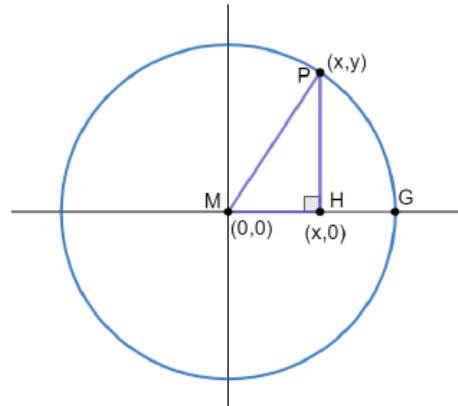
After the activity: What do you notice? What did you discover? How would you describe the relationship between a circle and its equation? How is that represented with its graph?



As 12.7.2 Exploration is meant to derive the equation of the circle, it is not necessary for the students to determine the actual length.

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

2. The center of circle M is located at $(0,0)$. Point P lies on circle M at the point (x,y) . Right triangle MPH is drawn inside the circle, such that $\overline{MP}$ is a circle's radius and point H , $(x,0)$, lies on radius $\overline{MG}$.



How is this circle different from the circle in question 1? How is finding the distance of each line segment similar? How is it different?

- a. Write an expression that represents the length of each side.

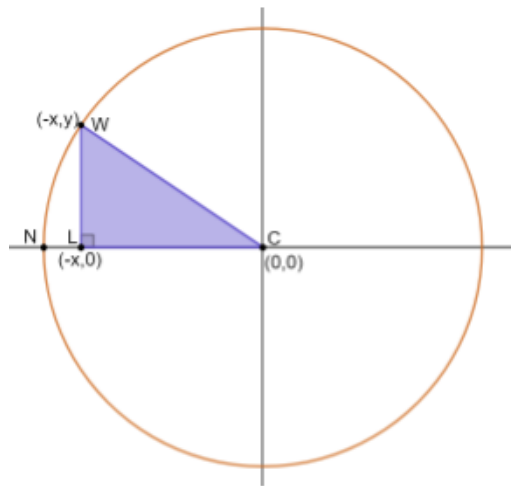
Line Segment	Distance Between Endpoints
$\overline{MH}$	x
$\overline{HP}$	y

- b. Use the expressions you found for the lengths of $\overline{MH}$ and $\overline{HP}$ to complete the equation that can be used to determine the length of $\overline{MP}$ by applying the Pythagorean Theorem.

$$(MP)^2 = (\quad x \quad)^2 + (\quad y \quad)^2$$

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

3. The center of circle C is located at $(0,0)$. Point W lies on circle C at the point $(-x,y)$. Right triangle WLC is drawn inside the circle, such that WC is a circle's radius and point L , $(-x,0)$, lies on radius NC .



Students may need to be reminded that distance is always positive. Consider prompting students to engage the class in a discussion:

Can distance be negative? Why or why not?

- a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
LC	x
LW	y

- b. Use the expressions you found for the lengths of LC and LW to complete the equation that can be used to determine the length of WC by applying the Pythagorean Theorem.

$$(WC)^2 = (x)^2 + (y)^2$$

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

- c. Complete the table with the equation you found that can be used to determine the length of the radius of each circle.

Line Segment	Distance
$\overline{MP}$ radius of circle M	$(MP)^2 = x^2 + y^2$ or $(r)^2 = x^2 + y^2$
$\overline{WC}$ radius of circle C	$(WC)^2 = x^2 + y^2$ or $(r)^2 = x^2 + y^2$

In circle M and circle C , the location of the point on the circle was in a different location, but the equations that can be used to determine the length of the radius of the circles are the same.

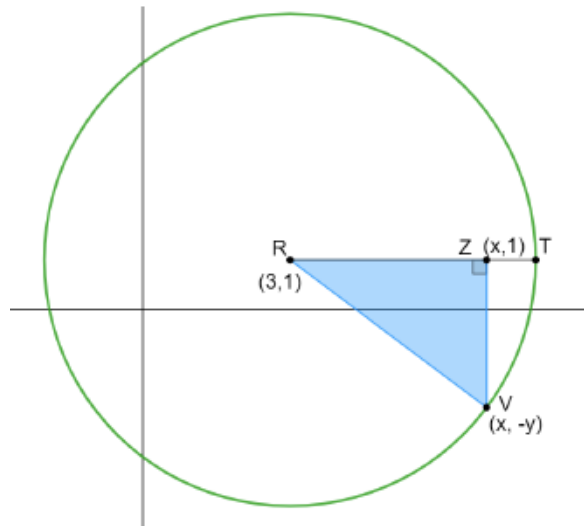


Use inquiry to drive learning. Listen for students who are stuck or students who need support making connections.

- What happens when you square a number?
- What happens when you square a negative number?
- Is there a difference between -5^2 and $(-5)^2$?
- Why is this difference important in deriving the equation of a circle?

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

4. The center of circle R is located at $(3,1)$. Point V lies on circle R at the point $(x, -y)$. Right triangle RZV is drawn inside the circle, such that $\overline{RV}$ is a circle's radius and point $Z, (x, 1)$, lies on radius $\overline{RT}$.



- a. Write an expression that represents the length of each side.

Line Segment	Distance Between Endpoints
$\overline{RZ}$	$x - 3$
$\overline{ZV}$	$1 + (-y)$

- b. Complete the equation that can be used to determine the length of the radius of circle R .
 $(r)^2 = (\quad x - 3 \quad)^2 + (\quad y - 1 \quad)^2$



Question 4 allows students to investigate equivalent forms of the circle. Consider investigating the different ways students can represent the distance and then explain why the value squared would be the same.

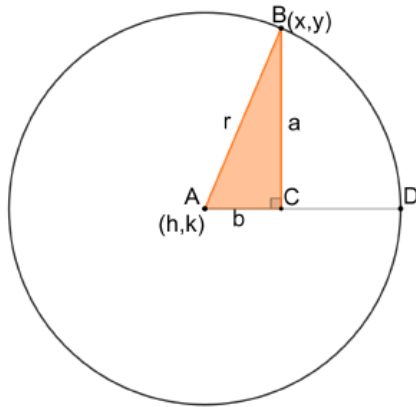


Question 4 provides an opportunity to determine students' understanding of the key elements of a circle and points in the coordinate plane.

Students who struggle with finding the distance in this problem might need you to provide numbers for some of the variables until they can see the relationship.

12.7.2 Exploration: Deriving the Equation of a Circle (continued)

5. The center of circle A is located at (h, k) . Point B lies on circle C at the point (x, y) . Right triangle ABC is drawn inside the circle such that $\overline{AB}$ is a circle's radius and point C , lies on radius $\overline{AD}$. $AC = b$, $BC = a$, and $AB = r$.



- Use (x, y) and (h, k) to determine the ordered pair for point C .
 (x, k)
- Fill out the table by using the ordered pairs to complete each equation.

Line Segment	Distance Between Endpoints
$\overline{AC}$	$b = x - h$
$\overline{BC}$	$a = y - k$

The Pythagorean theorem can be used to write the equation, $a^2 + b^2 = r^2$.

- Use the equations in the table to substitute for a and b in the equation $a^2 + b^2 = r^2$.
 $(x - h)^2 + (y - k)^2 = r^2$



The **center-radius form** of an equation of a circle with center, (h, k) , and radius, r , is $(x - h)^2 + (y - k)^2 = r^2$.



Students may struggle with question 5. Refer them to the other circles to determine the relationship.

- Why is the x -value of the ordered pair the same as point A ?
- Why is the y -value of the ordered pair the same as point B ?



Consider having students discuss what they have learned in 12.7.2 Exploration with a partner or small group using the Compare and Connect instructional routine. Students are encouraged to share comparisons found, make connections between each equation they wrote throughout this component, and discuss how the equation of a circle comes from it.

12.7.3 Guided Instruction: Key Features of a Circle

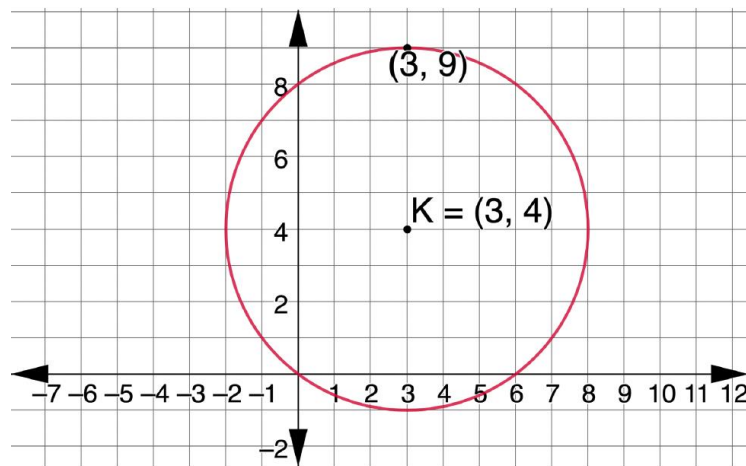
Component Overview: Students are using the equation of a circle derived in 12.7.3 Exploration to write the equation of a circle when given a center and radius.



12.7.3 Guided Instruction: Key Features of a Circle

Circles have two key features: the center and the radius. The key features can be used to write the equation, $(x - h)^2 + (y - k)^2 = r^2$, of the circle.

1. Circle K is shown with center $(3, 4)$. The point $(3, 9)$ is on circle K .



- a. Determine the length of circle K 's radius.
 $r = 5$
- b. Write the equation that represents circle K .
 $(x - 3)^2 + (y - 4)^2 = 25$ or
 $(x - 3)^2 + (y - 4)^2 = 5^2$

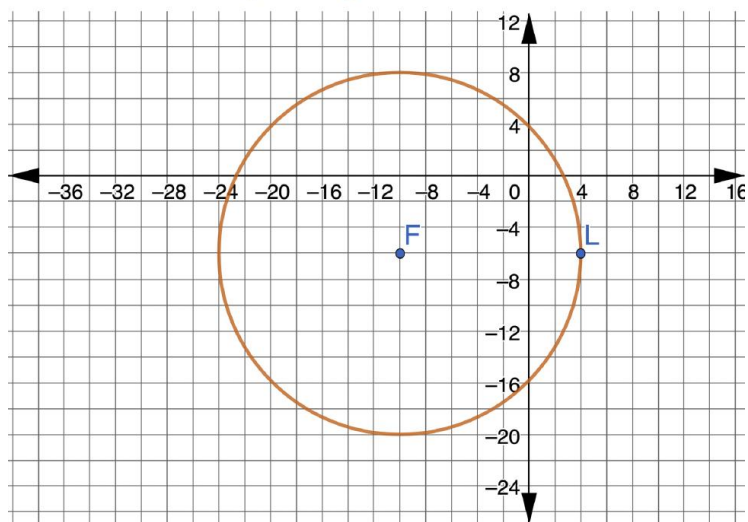


Notice and Wonder

Consider providing students the opportunity to write down things they notice and wonder about the equation of the circle and the circle provided in question 1. This will allow students to make connections between the equation and the information given.

12.7.3 Guided Instruction: Key Features of a Circle (continued)

2. Circle F is shown, where point L is on the circle.



Provide opportunity for synthesis using question 2. Students spent time exploring in the first half of this lesson, and the next activity is a formative Wrap-Up. Utilize question 2 to ask questions that position students as sensemakers to synthesize their learning to apply during 12.7.4 Wrap-Up.

- a. Determine the key features of circle F .

Key Features	
Radius	14
Center	$(-10, -6)$

- b. Write the equation that represents circle F .

$$(x + 10)^2 + (y + 6)^2 = 196 \text{ or}$$

$$(x + 10)^2 + (y + 6)^2 = 14^2$$

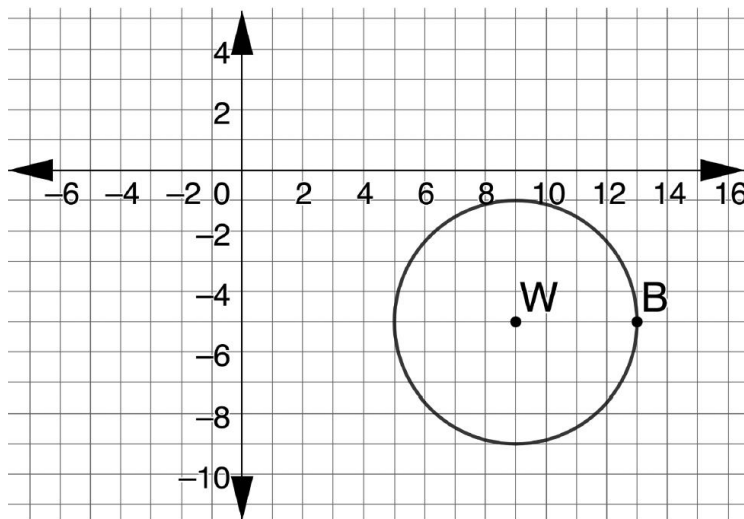
12.7.4 Wrap-Up: Ticket Out the Door

Component Overview: Students apply their knowledge of writing the equation of a circle, given a graph. Students write the key features of a circle, then write the equation of the circle.



12.7.4 Wrap-Up: Ticket Out the Door

1. Circle W is shown, where point B is on the circle.



- a. Determine the key features of circle W .

Key Features	
Radius	4
Center	(9, -5)

- b. Write the equation that represents circle W .

$$(x - 9)^2 + (y + 5)^2 = 16 \text{ or}$$

$$(x - 9)^2 + (y + 5)^2 = 4^2$$



Discuss any questions the students still might have regarding this lesson. Consider a Parking Lot approach with a list of the questions on the board or poster in which students can either answer each other's questions to replace a missing assignment grade, or you can revisit as a class. This will give you a quick way to assess how your students are understanding this lesson and its material.